Chapter 1

Introduction

1.1 Overview

With the increasing reliance on external storage devices such as USB drives, external hard disks, and SSDs, ensuring data security has become a major concern. Unauthorized access to these devices can lead to data breaches, information theft, and malware infections. The **Password Protected External Device Security System** is designed to provide an additional layer of security by restricting access to authorized users only. This system employs encryption techniques and authentication mechanisms to safeguard sensitive information. It ensures that only users with the correct credentials can access the data, making it ideal for personal, corporate, and governmental use. The integration of biometric authentication and cryptographic security further enhances protection, preventing unauthorized access even if the device is lost or stolen.

1.2 Aim/Motivation

The primary aim of this system is to enhance data security on external storage devices by implementing a password protection mechanism. The increasing number of cyber threats, data breaches, and unauthorized access cases highlight the need for a reliable security system. Many external devices lack strong security measures, making them vulnerable to hacking and data theft. This project is motivated by the necessity to safeguard confidential data from unauthorized users while ensuring ease of use for legitimate users. Additionally, the system aims to provide an adaptable and platform-independent security solution that can be seamlessly integrated into various operating systems.

1.3 Objective

The objectives of the **Password Protected External Device Security System** include:

- **Implementing a secure authentication mechanism** that requires users to enter a password before accessing the device.
- **Preventing unauthorized access** by implementing features like device lockdown or automatic data encryption after multiple incorrect password attempts.
- **Ensuring platform compatibility**, making the system usable across Windows, Linux, and macOS.

1.4 Organization of Report

This report is structured to provide a comprehensive understanding of the Password Protected External Device Security System:

- **Chapter 1 – Introduction:** Provides an overview, aim, motivation, objectives, and organization of the report.
- **Chapter 2 – Literature Review:** Discusses existing security mechanisms for external devices and their limitations.
- **Chapter 3 – Problem Statement:** External storage devices such as USB drives are highly vulnerable to unauthorized access, data theft, and malware attacks due to weak authentication mechanisms, necessitating a robust password protection system to secure sensitive information.
- **Chapter 4 – Software Requirement and Specification:** The system requires a user authentication module, encryption algorithms, cross-platform compatibility, and real-time monitoring features to ensure secure access and protection of external storage devices.
- **Chapter 5 – System Design:** The system design includes user authentication, encryption mechanisms, and access control to secure external storage devices against unauthorized access.
- **Chapter 6 – Conclusion and Future Scope:** Summarizes findings and suggests potential improvements for the system.
- **Chapter 7 – References:** Information gathering from various book, webpages, blogs for project building.

Chapter 2

Literature Review

The Password Protected External Device Security System builds upon existing research and technological advancements in data protection, authentication mechanisms, and encryption techniques. This chapter reviews various security strategies implemented in external storage devices and evaluates their effectiveness and limitations.

2.1 Existing Security Mechanisms

Several security mechanisms have been developed to protect external storage devices, including:

- Hardware Encryption: Some high-end USB drives and external hard drives come with built-in encryption chips that automatically encrypt stored data.
- Software Encryption: Various encryption tools such as BitLocker (Windows), VeraCrypt, and AxCrypt provide software-based encryption solutions for external devices.
- Password Protection Software: Some devices include built-in software that prompts users to enter a password before accessing the data.
- Biometric Authentication: Advanced security systems integrate fingerprint scanners and facial recognition to enhance security.
- Two-Factor Authentication (2FA): Some modern devices implement two-step verification using OTP (One-Time Password) or authentication apps.

2.2 Limitations of Existing Security Systems

Despite these advancements, several limitations exist in current security implementations:

- Vulnerability to Brute Force Attacks: Password-based systems are susceptible to brute force attacks if passwords are weak or not regularly updated.
- Device Compatibility Issues: Some encryption software may not be cross-platform compatible, limiting usability across different operating systems.
- User Convenience vs. Security Trade-off: Biometric authentication offers higher security but may introduce usability challenges if the sensor fails.

- Data Recovery Complexity: Encrypted external devices often make data recovery difficult if the password is forgotten or lost.
- High Implementation Costs: Hardware-based encryption solutions are more expensive compared to software-based security solutions.

2.3 Recent Advancements and Research

Recent research in the field of cybersecurity and data protection has introduced various techniques to enhance external storage security:

- AI-Powered Threat Detection: Machine learning algorithms analyze user behavior to detect and prevent unauthorized access attempts.
- Blockchain-Based Security: Decentralized security mechanisms using blockchain technology provide tamper-proof authentication and data integrity.
- Quantum Cryptography: Advanced encryption techniques based on quantum computing promise stronger security measures against cyber threats.
- Cloud-Based Authentication: Integration with cloud security services enables remote authentication and monitoring of external devices.

Chapter 3

Problem Statement

External devices such as USB drives/duckies which are being used extensively are vulnerable to cyber at-tacks either by spreading viruses or giving others access to certain data. The absence of a strict authentication mechanism is a weakness and thus becomes vulnerable to various breaches. As long as safeguarding mechanisms remain absent these systems are under threat of hacking leaks of data and virus infections with personal organizational systems.

Who: The problem affects individuals, businesses, and organizations that rely on external devices to store and transfer sensitive data.

What: The key issue is the lack of a robust security mechanism to protect external storage devices. The absence of strong authentication methods increases the likelihood of unauthorized data access and potential cyber threats.

When: The issue arises whenever external storage devices are used in unsecured environments, lost, or accessed by unauthorized users. It needs to be addressed immediately to prevent potential data theft.

Where: This issue occurs across various domains, including corporate environments, government agencies, and personal data storage systems where external devices are frequently used.

Why: Addressing this issue is crucial to prevent data breaches, unauthorized access, and information theft, which can have severe consequences, including financial losses, reputational damage, and legal liabilities.

We enhances traditional security measures by integrating password authentication, encryption, and real-time monitoring. Our project further strengthens this approach by introducing an additional safety layer, adding an extra level of precaution to prevent unauthorized access and ensure robust data protection for external storage devices

External Hardware Risks to System Security

1.O.MG ELITE

The O.MG ELITE is a powerful and covert hacking device disguised as an ordinary USB cable, but inside it hides advanced components capable of launching cyberattacks. Once plugged into a computer, laptop, or smartphone, it can remotely inject keystrokes, run malicious commands, steal sensitive data, and even create backdoors—all without the user noticing. It poses a serious threat to individuals, companies, and organizations by turning a simple-looking cable into a weapon for cybercriminals. People who often borrow or use public charging cables are especially at risk, making awareness and caution extremely important in today's digital world.

2.USB Rubber Ducky

The USB Rubber Ducky is a deceptive and dangerous hacking tool that looks like a regular USB flash drive but acts like a keyboard when plugged into a device. It can automatically type and execute pre-programmed commands at lightning speed, allowing attackers to install malware, steal credentials, disable security settings, or gain unauthorized access to systems. Once connected to a PC, laptop, or any USB-enabled device, it can compromise the system within seconds. This tool is often used in penetration testing but can be misused by cybercriminals, posing a serious threat to anyone who plugs in unknown or suspicious USB devices.

3. Computer mouse

A regular-looking computer mouse can be secretly modified into a hacking device, turning it into a serious cyber threat. These weaponized mice may still function normally for clicking and scrolling but can also act like a keyboard to inject malicious commands once plugged in. This type of attack works similarly to a USB Rubber Ducky, allowing hackers to steal data, install malware, or gain remote access to a system—often in just a few seconds. Because it looks like an ordinary device, users are less likely to suspect it, making it a stealthy tool used in physical access attacks. Any PC, laptop, or system where such a device is connected can be at serious risk.

4.A keyboard

A keyboard can also be modified and used as a hacking device, making it a silent threat in disguise. While it appears and functions like a normal keyboard, it can be programmed to automatically inject malicious commands as soon as it’s connected to a computer. Just like the USB Rubber Ducky, these weaponized keyboards act as Human Interface Devices (HIDs), giving attackers the ability to install malware, steal sensitive information, disable security settings, or open backdoors within seconds. Because users trust everyday peripherals like keyboards, this method is highly effective in physical attacks, especially in offices, schools, or shared environments where devices can be easily swapped or tampered with.

Chapter 4

Hardware and Software Requirements Specification

4.1 Hardware Requirements

1. **Personal Computer or Laptop with USB Ports**

- A computer or laptop with **functional USB ports** is essential for connecting external storage devices.
- USB **3.0 or higher** is recommended for faster data transfer rates.

2. **External Storage Device .**

- The security system should be compatible with multiple types of external storage, including:
    1. **USB Flash Drives** – commonly used for portable data storage.
- The system will implement **authentication** mechanisms to prevent unauthorized access.

4.2 Software Requirements

3. **Operating System: Windows, Linux, or macOS**

- The system should be **platform-independent**, working seamlessly on:
    1. **Windows (10/11)** – One of the most widely used OS, supporting a variety of security applications.
    2. **Linux (Ubuntu, Fedora, Kali Linux, etc.)** – Preferred for cybersecurity and encryption applications.
    3. **macOS** – Known for its built-in security and encryption tools.
- The software should be **compatible with all file systems** (NTFS, FAT32, exFAT, APFS, etc.).

4. **Programming Languages**

    1. **Python:**
        - Used for implementing and authentication processes.
        - Flask for primary backend framework.
    2. **CSS & JavaScript:**
        - Used for designing a user-friendly **Graphical User Interface (GUI)**.
        - JavaScript can also help with **frontend encryption and validation**.
    3. **Optional:**
        - **C++ or Java** for system-level programming if performance optimization is needed.

Chapter 5

System Design

5.1 Project Block Diagram

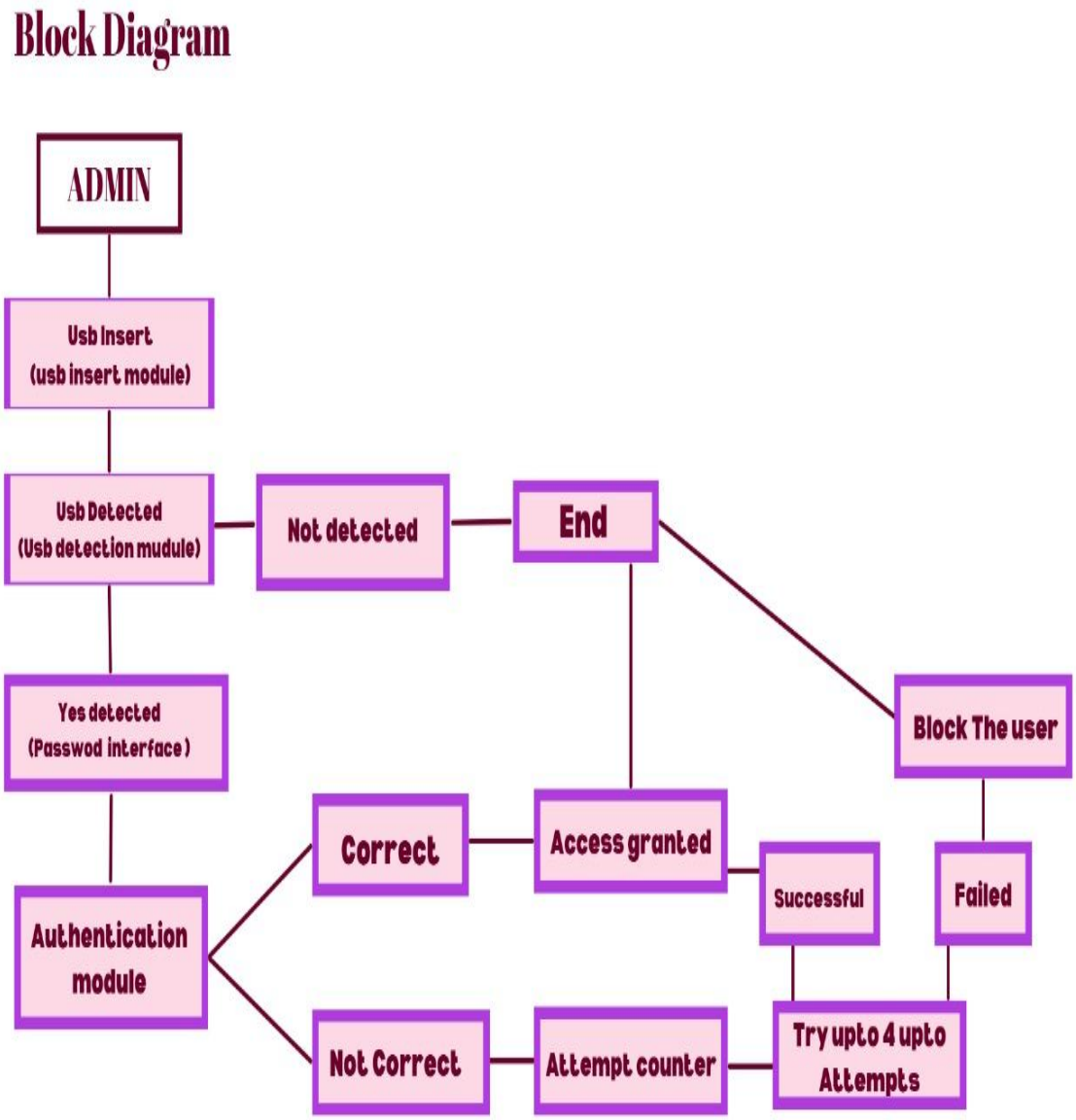

Figure 1.1 Block Diagram for Password Protected External Device Security System

Block Diagram Explanation
The block diagram represents the logical flow of operations for the **Password Protected External Device Security System**. It outlines the steps taken from the insertion of a USB device to granting or denying access based on authentication.

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1. Admin
The process begins under the control of the system administrator. The administrator is responsible for monitoring and managing access to the external device.

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2. USB Insert (USB Insert Module)
This module detects the physical insertion of a USB device into the system. It initiates the further process of validation and authentication.

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3. USB Detected (USB Detection Module)
After insertion, the USB Detection Module checks whether a USB device has been successfully detected by the system.
If the USB is not detected, the process is terminated, and the system proceeds to the **End** state.
If the USB is detected, the system advances to the password verification stage.

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4. Password Interface
Upon successful detection of the USB, the system prompts the user to enter a password. This interface allows the user to input credentials for authentication.

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5. Authentication Module
The Authentication Module is responsible for verifying the password entered by the user.
If the password is correct, access to the USB device is granted.
If the password is incorrect, the system initiates a counter to track the number of failed attempts.

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6. Attempt Counter
The system allows the user up to four attempts to enter the correct password. This is a security measure to prevent unauthorized access.
If the correct password is entered within four attempts, the system proceeds to **Access Granted**.
If all four attempts fail, the system proceeds to **Block the User**.

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7. Access Granted
When the user provides the correct password within the allowed number of attempts, full access to the USB device is granted.

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8. Block the User

If the user fails to provide the correct password in all four attempts, the system blocks the user to prevent further access. This serves as a security mechanism against brute-force attacks or unauthorized usage.

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9. End

The process concludes in any of the following scenarios:

5.2 Flowchart Diagram

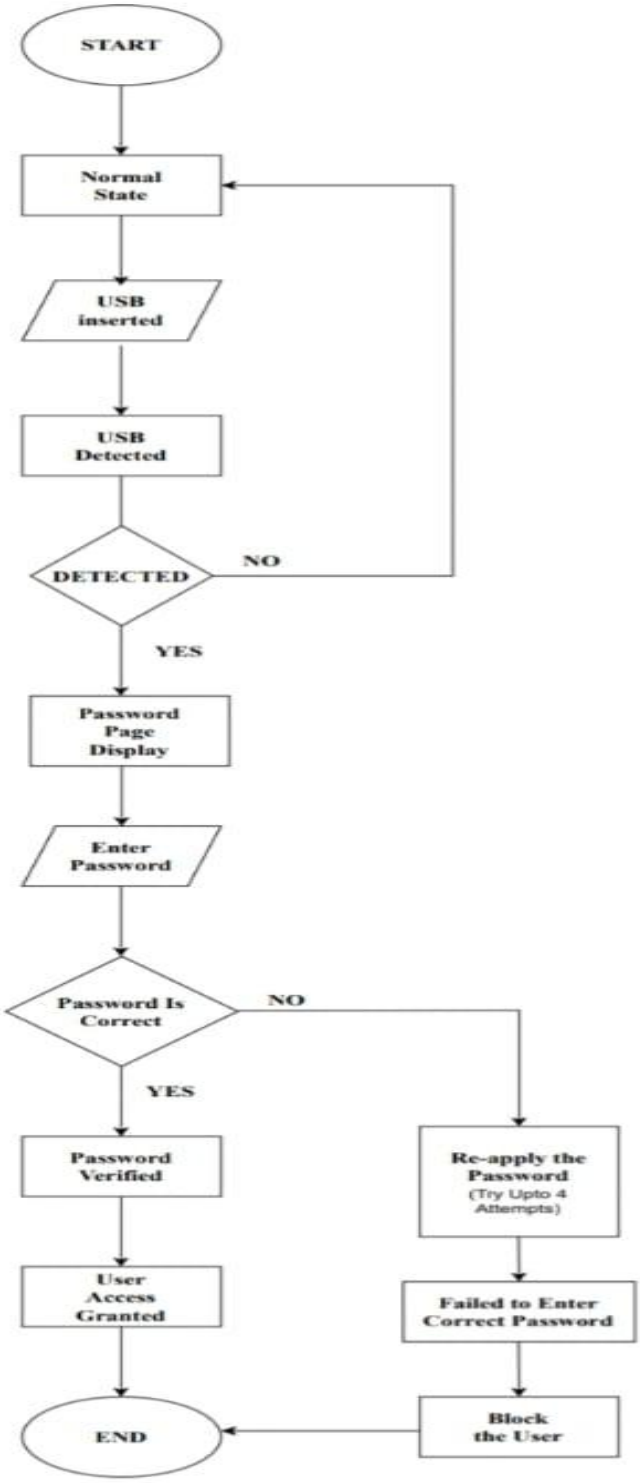

Fig 2.1 Process Flowchart of Password Protected External Device Security System

The flowchart illustrates the **Password Protection System for External Storage Devices** by outlining the authentication process when a USB device is inserted.

Explanation of the Flowchart:

1. **Start:** The system is in a normal state before any external device is inserted.
2. **USB Inserted:** When a USB device is connected, the system detects it.
3. **USB Detected:**
   - If the USB is not detected, the system returns to the normal state.
   - If the USB is detected, the system proceeds to authentication.
4. **Password Prompt:** A password entry screen is displayed, prompting the user to input credentials.
5. **Password Verification:**
   - If the password is correct, access is granted, and the user can access the storage device.
   - If the password is incorrect, the system allows up to four attempts.
6. **Failure Handling:**
   - If the correct password is not entered within four attempts, the system blocks the user to prevent unauthorized access.
7. **End:** The process concludes when access is either granted or denied.